SYNOPSIS

Objective of the Analysis

This paper examines the root causes of cost overruns in investment projects with the aim of identifying, grouping, and quantifying the most significant risk factors. The dataset represents the opinions of 143 construction industry professionals such clients, consultants, and contractors who rated 44 potential cost overrun factors on a five-point scale (1=least significant, 5=most significant). The ultimate goal was to build a risk assessment and forecasting model at the individual respondent level, as well as segment respondents into meaningful risk groups for practical project management.

Analytical Methods Used

1) The first step was data preparation and descriptive analysis. The dataset was cleaned in three iterations: at each step, variables that did not meet the assumptions of the multivariate analysis were removed. Ultimately, the working dataset included 44 variables. Descriptive statistics were calculated for all variables: mean, standard deviation, skewness, and kurtosis. Factor means ranged from approximately 2.49 to 4.07, indicating that respondents rated most factors as moderately or highly significant.

2) The second step of our study was an exploratory factor analysis, conducted using the principal component analysis method and Varimax rotation. For this, we calculated the KMO measure of sampling adequacy, the full correlation matrix, and its inverse. An eigenvalue was also obtained, and a scree plot was constructed to optimally select it. Six factors with eigenvalues greater than one were selected, collectively explaining approximately 62.6% of the variance, with the first factor alone explaining 29.4%.

3) The third step involved understanding and naming the factors. Based on the structure of the factor loadings, each factor was given a conceptual name reflecting the factor's underlying nature. All six factors:

- Factor 1, Construction Quality and Facility Management Issues: Associated with low labor productivity (-0.862), poor construction quality (-0.812), construction errors (-0.806), and lack of engineering experience (-0.791).
- Factor 2, Designers' Violations in Design and Project Plans: Explained by deficiencies in design documentation (0.656); deficiencies in working drawings (0.642); and incorrect calculation of work volumes (0.655).
- Factor 3, Financial and Administrative Delays: Exacerbated by delays in payment for completed work (-0.753); cash flow problems (-0.707); contractor financial problems (-0.619); and delays by authorized bodies (-0.639).
- Factor 4, Changes in Scope and Customer Requirements: Includes changes in scope and specifications (0.795), changes in customer requirements (0.769) and frequency of additional work (0.763).

- Factor 5, Supply and Procurement Problems: is associated with monopoly of material suppliers (-0.762), delivery delays (-0.617) and limited range of materials on the local market (-0.618).
- Factor 6, Contract Management and External Conditions: is caused by lack of time for preparation of contract documentation (-0.663), weather conditions (-0.667).

4) **The fourth step** was to assess the factors, which entailed conducting regression analysis to establish a value for each instance (project/respondent) over the defined range of parameters. Using these assessments as inputs of a multiple regression model, a new regression outcome was produced, which showed the contribution of each element to the total project risk.

5) **The fifth step** includes creating a risk score. This score captures the weighted impact of all of the factors and facilitates the categorization of projects into risk categories (from very low risk to very high risk). This step represents a standard model for risk quantification and ranking that is commonplace in operations management and decision analytics.

6) **The sixth step** included analysis of segmentation to cluster the projects by their risk profiles and other attributes, including geography, and level of experience. This provides helpful details such as pinpointing high-risk environments or highlighting specific patterns tied to organizational or geographical variations.